CheckMySmile – AI Dental Health Screening

Concept Note

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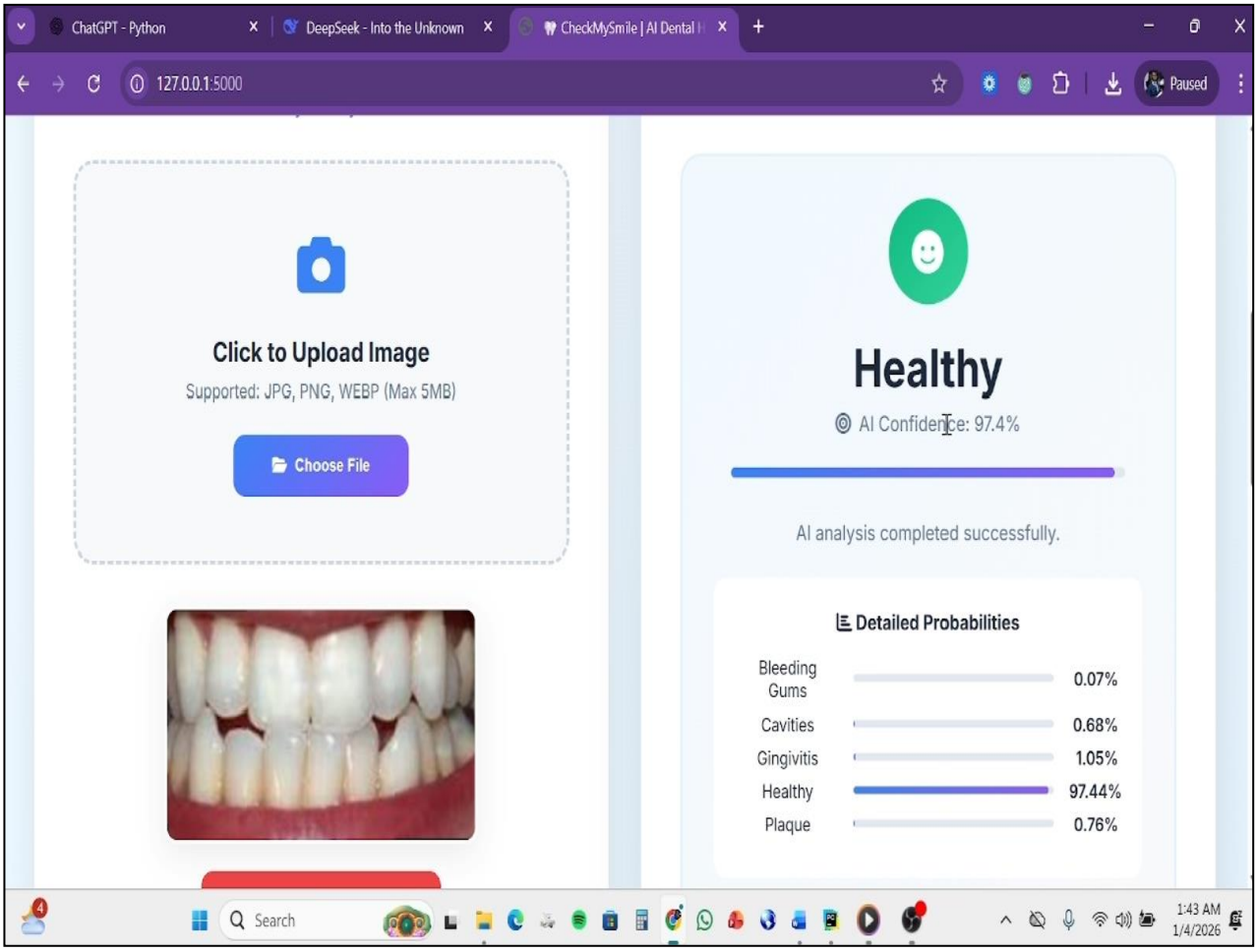
Executive Summary

CheckMySmile is a web-based platform that uses AI to provide preliminary dental screening and promote preventive oral health. Users take photos of their teeth, and the system highlights potential issues such as cavities, gum inflammation, and plaque buildup.

The innovation targets students, low-income families, and underserved communities with limited access to dental care. By combining easy-to-use technology with educational guidance, CheckMySmile empowers individuals to monitor oral health, seek timely professional care, and adopt better hygiene habits.

This student-led innovation is currently piloted locally, with plans to scale regionally and nationally, providing an affordable and accessible digital health solution for Malawi.



Problem Statement

Oral health challenges are significant in Malawi:

- Many people delay dental visits due to high costs, distance to clinics, or fear of treatment.
- There is a low dentist-to-population ratio, leading to late detection of preventable dental conditions.
- Lack of public awareness about early signs of oral disease results in advanced cases that are painful and expensive to treat.

A rural area in Malawi. Oral health local

Locals accessing oral treatment in a Local setting somewhere in Malawi

CheckMySmile addresses these challenges by providing early, accessible, and practical dental screening, empowering communities to act before problems escalate.

Innovation Description

- CheckMySmile operates as a web-based platform using AI image analysis

- Users take clear photos of their teeth and gums following on-screen guidance.

- The images are processed to enhance lighting, clarity, and detect patterns like discoloration, plaque, or gum redness.

- A trained AI model identifies potential dental issues and provides risk-based feedback (low, moderate, high) along with actionable guidance.

- Additional features include educational resources, reminders for checkups, and secure tracking of previous scans. The platform is designed to work on low-resource systems and is currently running on a local server.

Unique Value Proposition

CheckMySmile stands out because:

- It is locally developed in Malawi for the specific context of students and underserved communities.
- Combines early detection, education, and practical guidance in one accessible tool.
- Unlike traditional dental apps, it does not require expensive equipment or clinic visits.
- Designed to be low-cost, easy-to-use, and scalable for wider adoption.

Target Users & Beneficiaries

Primary Users:

- Students and young adults in universities and schools.
- Low-income families and individuals in rural areas.

Secondary Users / Beneficiaries:

- Children learning preventive oral habits.
- Dental professionals and public health systems, benefiting from early detection and reduced clinic burden.

Implementation & Deployment

Current Deployment:

- Community-level pilot at Kamuzu University of Health Sciences, running on a local server.

Scaling Plan:

- **Short-term**: Expand to other schools and communities in Blantyre.
- **Medium-term:** Scale nationally across Malawi through partnerships with health organizations and schools.
- **Long-term:** Potential regional adoption in neighboring countries with similar challenges.

Social & Economic Impact

Social Impact:

- Improves oral health awareness and hygiene habits.
- Encourages proactive dental care, reducing pain and complications.

Economic Impact:

- Reduces treatment costs by detecting issues early.
- Decreases strain on clinics and public health facilities.
- Empowers communities with digital health literacy.

Technical Approach & Sustainability

Technical Overview:

- Web-based platform using AI for image analysis.
- Currently running on a local server; stores data securely.
- Designed for low-resource settings with simple interfaces.

Sustainability / Future Model:

- Freemium approach: basic screening free, optional paid features for reports or advanced guidance.
- Open-source tools reduce maintenance costs.
- Potential collaborations with universities, health organizations, and schools.

Key Risks & Mitigation:

- Accuracy: Continuous testing and AI refinement.
- Access: Lightweight design and community deployment.
- Privacy: Local storage and secure handling, with future encryption for scaling.